

# WI-FI NETWORK AUDITING USING AIRCRACK-NG



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## 1. Introduction

Wi-Fi is commonly used to connect laptops, smartphones, and other devices to a network without cables. Since the communication happens through wireless signals, the security of the network needs to be checked regularly.

For this project, I performed a basic Wi-Fi security audit using Kali Linux and the Aircrack-ng toolset. The testing was carried out only on an authorized test network in a controlled laboratory environment.

The project involved checking the Wi-Fi adapter, checking the wireless interface, enabling monitor mode, scanning nearby wireless networks, identifying the authorized test network, observing a connected test device, and carrying out a controlled password-strength test using captured authentication data.

The main purpose was to understand the basic process of a Wi-Fi security audit and to see why strong Wi-Fi security settings and passwords are important.

## 2. Project Background and Problem Statement

### Project Background

Wi-Fi networks are used every day to connect computers, phones, and other devices. Because wireless communication travels through radio signals, it is useful to check whether the network is properly protected.

Security problems can occur because of a weak password, outdated router software, or incorrect security settings. A security audit can help identify these weaknesses so that they can be fixed.

This project focuses on learning the basic process of Wi-Fi security auditing with Kali Linux and Aircrack-ng. The practical work was done on an authorized test network in a controlled environment.

### **Problem Statement**

Some Wi-Fi networks may have weak security settings or weak passwords. If these weaknesses are not identified, the network can become easier to attack.

Therefore, this project was carried out to understand how a Wi-Fi network can be checked safely in an authorized environment. The work focuses on wireless monitoring, network scanning, WPA/WPA2 authentication, targeted packet capture, and controlled password-strength testing.

The project does not include testing networks without permission or accessing devices that are not part of the authorized environment.

## 3. Aim and Objectives

### Aim

The main aim of this project is to learn and demonstrate the basic process of auditing an authorized Wi-Fi network using Aircrack-ng on Kali Linux.

### Objectives

- Check whether the USB Wi-Fi adapter is detected properly.
- Check the wireless interface.
- Enable monitor mode on the Wi-Fi adapter.
- Scan the available wireless networks.
- Identify the authorized test access point.
- Record the SSID, BSSID, channel, and security information.
- Observe the controlled client connected to the test network.
- Perform a targeted packet capture.
- Understand WPA/WPA2 authentication.
- Understand the purpose of a WPA/WPA2 handshake.
- Perform a controlled dictionary-based password-strength test.
- Record the results and findings.
- Provide basic security recommendations.
- Understand the legal and ethical requirements of Wi-Fi testing.

## 4. Scope

This project is limited to basic Wi-Fi security testing in an authorized laboratory environment.

The project covers:

- Kali Linux testing environment
- USB Wi-Fi adapter setup
- Wireless interface checking
- Monitor mode
- Wireless network scanning
- Authorized packet capture
- WPA/WPA2 testing
- Controlled password-strength testing
- Documentation of findings
- Security recommendations
- Legal and ethical aspects of wireless testing

The project does not include unauthorized access, credential theft, malware, persistence, or testing networks without permission.

## 5. Hardware and Software Requirements

| Requirement                        | Purpose                        |
|------------------------------------|--------------------------------|
| Laptop/PC                          | To run Kali Linux              |
| TP-Link 802.11ac USB Wi-Fi Adapter | To perform wireless monitoring |
| Wi-Fi Router/Access Point          | Authorized test network        |
| Test Phone/Laptop                  | Controlled Wi-Fi client        |
| Kali Linux                         | Testing environment            |

|                   |                             |
|-------------------|-----------------------------|
| Aircrack-ng Suite | Wireless security auditing  |
| iwconfig          | Wireless interface checking |
| ip link           | Interface verification      |

The practical setup used Kali Linux with a TP-Link 802.11ac USB Wi-Fi adapter. The wireless interface used during the testing was wlan0.

## 6. Basic Wi-Fi Terms

| Term         | Simple Meaning                                                                            |
|--------------|-------------------------------------------------------------------------------------------|
| SSID         | The name of a Wi-Fi network.                                                              |
| BSSID        | The MAC address used to identify a wireless access point.                                 |
| Channel      | The radio channel used by a Wi-Fi network.                                                |
| WPA/WPA2     | Security methods used to protect Wi-Fi networks.                                          |
| CCMP         | An encryption method commonly used with WPA2.                                             |
| PSK          | Pre-shared key authentication commonly used in home Wi-Fi networks.                       |
| Handshake    | Authentication messages exchanged when a Wi-Fi client connects.                           |
| Monitor Mode | A mode that allows a Wi-Fi adapter to observe wireless traffic during authorized testing. |

These terms help explain the wireless information collected during the project.

## 7. Tools Used

The following tools were used or documented during the project.

### airmon-ng

It was used to check the wireless adapter and enable monitor mode.

### airodump-ng

It was used to scan wireless networks and observe wireless information.

### aircrack-ng

It was used to test password candidates against authorized captured authentication data.

```
iwconfig
```

It was used to check the wireless interface and its mode.

```
ip link
```

It was used to confirm that the wireless interface was available in Kali Linux.

## 8. Experimental Setup

The practical setup consisted of Kali Linux, a TP-Link 802.11ac USB Wi-Fi adapter, an authorized Wi-Fi access point, and a controlled client device.

The wireless interface used was wlan0.

The test access point identified during the project was TP-Link\_6922.

The wireless security information observed was:

- SSID: TP-Link\_6922
- BSSID: B8:FB:B3:1C:69:22
- Channel: 2
- Security: WPA2
- Cipher: CCMP
- Authentication: PSK

A controlled phone or laptop was used as the client device.

## 8.1 Kali Linux Live Mode

The project instructions mention using Kali Linux in Live Mode.

Live Mode allows Kali Linux to run from a bootable USB without permanently installing it on the computer. After starting Kali Linux, the compatible USB Wi-Fi adapter can be connected and checked before starting the wireless audit.

## 9. Methodology

The practical work was completed through a series of basic steps.

### Step 1 – Check the Wi-Fi Adapter

```
iwconfig
```

*Checks whether the wireless adapter and wlan0 interface are detected.*

First, the USB Wi-Fi adapter was connected to the computer.

The wireless interface wlan0 was detected successfully.

Result: The Wi-Fi adapter was recognized by Kali Linux.

```
kali@kali:~ % iwconfig
lo          no wireless extensions.

eth0        no wireless extensions.

wlan0       IEEE 802.11  ESSID:"TP-Link_6922"
            Mode:Managed  Frequency:2.417 GHz  Access Point: B8:FB:B3:1C:69:22
            Bit Rate=216 Mb/s   Tx-Power=20 dBm
            Retry short limit:7   RTS thr:off   Fragment thr:off
            Power Management:on
            Link Quality=70/70   Signal level=-34 dBm
            Rx invalid nwid:0   Rx invalid crypt:0   Rx invalid frag:0
            Tx excessive retries:0   Invalid misc:1   Missed beacon:0
```

Figure 1. Wi-Fi adapter and wlan0 interface detected using iwconfig.

### Step 2 – Check the Wireless Interface

```
ip link
```

*Verifies that wlan0 is available.*

The wireless interface was checked to make sure Linux had created the required interface.

Result: The wlan0 interface was available.

```
kali@kali:~ % ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:e9:1c:bf brd ff:ff:ff:ff:ff:ff
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DORMANT group default qlen 1000
    link/ether d4:d6:df:3f:8a:5d brd ff:ff:ff:ff:ff:ff
```

Figure 2. Wireless interface wlan0 shown using ip link.

### Step 3 – Check the Adapter

```
sudo airmon-ng
```

*Shows the wireless interface, driver, and chipset information.*

The Aircrack-ng tools were used to identify the wireless adapter, driver, and chipset.

The adapter was identified as a TP-Link 802.11ac NIC using the rtl88\_8822bu driver.

```
kali@kali:~ % sudo airmon-ng
[sudo] password for kali:

PHY      Interface      Driver      Chipset
phy1     wlan0          rtw88_8822bu  TP-Link 802.11ac NIC
```

Figure 3. TP-Link 802.11ac NIC, rtl88\_8822bu driver, and chipset information.

### Step 4 – Enable Monitor Mode

```
sudo airmon-ng start wlan0
```

*Enables monitor mode on the authorized test adapter.*

Monitor mode was enabled on the wireless adapter.

The interface name shown by Kali Linux was checked before continuing.

Result: Monitor mode was successfully enabled.

```
kali@kali:~ % sudo airmon-ng start wlan0

Found 2 processes that could cause trouble.
Kill them using 'airmon-ng check kill' before putting
the card in monitor mode, they will interfere by changing channels
and sometimes putting the interface back in managed mode

    PID Name
    539 NetworkManager
    911 wpa_supplicant

PHY      Interface      Driver      Chipset
phy1     wlan0          rtw88_8822bu  TP-Link 802.11ac NIC
                (monitor mode enabled)
```

Figure 4. Monitor mode successfully enabled on wlan0.

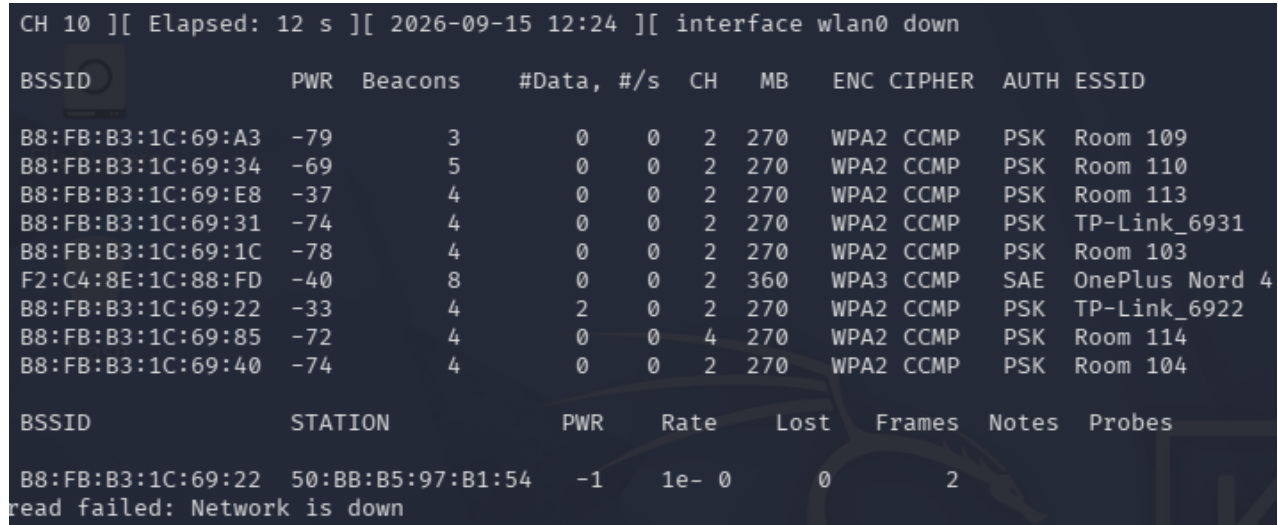
## Step 5 – Scan Wireless Networks

```
sudo airodump-ng wlan0
```

*Scans wireless networks in the authorized laboratory environment.*

The wireless environment was scanned to identify nearby Wi-Fi networks.

The authorized test network TP-Link\_6922 was visible during the scan.



| BSSID             | PWR | Beacons | #Data, #/s | CH | MB  | ENC  | CIPHER | AUTH | ESSID          |
|-------------------|-----|---------|------------|----|-----|------|--------|------|----------------|
| B8:FB:B3:1C:69:A3 | -79 | 3       | 0 0        | 2  | 270 | WPA2 | CCMP   | PSK  | Room 109       |
| B8:FB:B3:1C:69:34 | -69 | 5       | 0 0        | 2  | 270 | WPA2 | CCMP   | PSK  | Room 110       |
| B8:FB:B3:1C:69:E8 | -37 | 4       | 0 0        | 2  | 270 | WPA2 | CCMP   | PSK  | Room 113       |
| B8:FB:B3:1C:69:31 | -74 | 4       | 0 0        | 2  | 270 | WPA2 | CCMP   | PSK  | TP-Link_6931   |
| B8:FB:B3:1C:69:1C | -78 | 4       | 0 0        | 2  | 270 | WPA2 | CCMP   | PSK  | Room 103       |
| F2:C4:8E:1C:88:FD | -40 | 8       | 0 0        | 2  | 360 | WPA3 | CCMP   | SAE  | OnePlus Nord 4 |
| B8:FB:B3:1C:69:22 | -33 | 4       | 2 0        | 2  | 270 | WPA2 | CCMP   | PSK  | TP-Link_6922   |
| B8:FB:B3:1C:69:85 | -72 | 4       | 0 0        | 4  | 270 | WPA2 | CCMP   | PSK  | Room 114       |
| B8:FB:B3:1C:69:40 | -74 | 4       | 0 0        | 2  | 270 | WPA2 | CCMP   | PSK  | Room 104       |

| BSSID             | STATION           | PWR | Rate  | Lost | Frames | Notes | Probes |
|-------------------|-------------------|-----|-------|------|--------|-------|--------|
| B8:FB:B3:1C:69:22 | 50:BB:B5:97:B1:54 | -1  | 1e- 0 | 0    | 2      |       |        |

read failed: Network is down

Figure 5. Wireless scan showing the authorized TP-Link\_6922 access point and visible station.

## Step 6 – Record Target Information

The following information was recorded:

- SSID: TP-Link\_6922
- BSSID: B8:FB:B3:1C:69:22
- Channel: 2
- Security: WPA2
- Cipher: CCMP
- Authentication: PSK

## Step 7 – Start Targeted Capture

```
sudo airodump-ng --bssid B8:FB:B3:1C:69:22 --channel 2 wlan0
```

*Targets the authorized TP-Link\_6922 access point on channel 2.*

The capture was focused on the authorized test access point while the controlled client was connected or reconnecting.

## Step 8 – Observe the Client

The access point and client stations were visible during the targeted capture.

This confirmed that the controlled client could be observed in the test environment.

```
CH 2 ][ Elapsed: 24 s ][ 2026-09-15 12:30
```

| BSSID             | PWR | RXQ | Beacons | #Data, #/s | CH | MB  | ENC  | CIPHER | AUTH | ESSID        |
|-------------------|-----|-----|---------|------------|----|-----|------|--------|------|--------------|
| B8:FB:B3:1C:69:22 | -35 | 47  | 138     | 10 0       | 2  | 270 | WPA2 | CCMP   | PSK  | TP-Link_6922 |

| BSSID             | STATION           | PWR | Rate   | Lost | Frames | Notes | Probes |
|-------------------|-------------------|-----|--------|------|--------|-------|--------|
| B8:FB:B3:1C:69:22 | 50:BB:B5:97:B1:54 | -34 | 1e- 1  | 0    | 2      |       |        |
| B8:FB:B3:1C:69:22 | 86:EE:BE:3A:E0:69 | -42 | 0 - 6e | 0    | 9      |       |        |

Quitting...

Figure 6. Targeted capture showing TP-Link\_6922 and connected client stations.

## Step 9 – Capture Authentication Data

The capture was kept running while the controlled client reconnected.

A clear screenshot showing the WPA/WPA2 handshake indicator is now included in Figure 8. The screenshot shows “WPA handshake: B8:FB:B3:1C:69:22” while monitoring the authorized test network TP-Link\_6922 on channel 2.

The screenshot shows a terminal window titled "Shell No. 1" with a menu bar (Session, Actions, Edit, View, Help) and tabs for Shell No. 1, 2, and 3. The terminal output shows a capture on channel 2, elapsed 2 minutes, at 2026-09-15 12:30. A message "WPA handshake: B8:FB:B3:1C:69:22" is displayed. Below this is a table of detected networks, with the first row (B8:FB:B3:1C:69:22) highlighted by a red box. This is followed by a table of connected stations for the selected BSSID. The terminal ends with "Quitting..." and a prompt "kali@kali:~\$".

```
CH 2 ][ Elapsed: 2 mins ][ 2026-09-15 12:30 ][ WPA handshake: B8:FB:B3:1C:69:22
```

| BSSID             | PWR | RXQ | Beacons | #Data, #/s | CH | MB  | ENC  | CIPHER | AUTH | ESSID          |
|-------------------|-----|-----|---------|------------|----|-----|------|--------|------|----------------|
| B8:FB:B3:1C:69:22 | -35 | 47  | 138     | 10 0       | 2  | 270 | WPA2 | CCMP   | PSK  | TP-Link_6922   |
| 34:98:B5:BD:8D:C3 | -79 | 3   | 1152    | 1808 7     | 1  | 360 | WPA2 | CCMP   | PSK  | Room 109       |
| 34:98:B5:BD:8D:34 | -69 | 5   | 924     | 203 11     | 1  | 360 | WPA2 | CCMP   | PSK  | Room 110       |
| 34:98:B5:BD:8D:E8 | -37 | 4   | 418     | 102 2      | 1  | 270 | WPA2 | CCMP   | PSK  | Room 113       |
| 34:98:B5:BD:8D:31 | -74 | 4   | 356     | 78 6       | 1  | 270 | WPA2 | CCMP   | PSK  | TP-Link_6931   |
| 34:98:B5:BD:8D:1C | -78 | 4   | 299     | 51 11      | 1  | 270 | WPA2 | CCMP   | PSK  | Room 103       |
| F2:C4:8E:1C:88:FD | -40 | 8   | 276     | 90 6       | 1  | 360 | WPA3 | CCMP   | SAE  | OnePlus Nord 4 |
| B8:FB:B3:1C:69:85 | -72 | 4   | 184     | 36 4       | 1  | 270 | WPA2 | CCMP   | PSK  | Room 114       |
| B8:FB:B3:1C:69:40 | -74 | 4   | 162     | 24 6       | 1  | 270 | WPA2 | CCMP   | PSK  | Room 104       |

| BSSID             | STATION           | PWR | Rate   | Lost | Frames | Notes | Probes |
|-------------------|-------------------|-----|--------|------|--------|-------|--------|
| B8:FB:B3:1C:69:22 | 50:BB:B5:97:B1:54 | -34 | 1e- 1  | 0    | 2      |       |        |
| B8:FB:B3:1C:69:22 | 86:EE:BE:3A:E0:69 | -42 | 0 - 6e | 0    | 9      |       |        |

Quitting...

kali@kali:~\$

Figure 8. WPA handshake captured for the authorized test network TP-Link\_6922 (BSSID B8:FB:B3:1C:69:22) on channel 2.

## Step 10 – Password-Strength Test

```
sudo aircrack-ng -w <AUTHORIZED_WORDLIST> <CAPTURE_FILE>.cap
```

*Runs the password-strength test against authorized captured authentication data. Replace the placeholders with the approved laboratory files.*

Aircrack-ng was used with authorized captured authentication data and password candidates.

The supplied evidence shows Aircrack-ng testing password candidates and a later KEY FOUND result.

The actual password has not been included in the report.

```
Aircrack-ng 1.7
[00:00:17] 6933/100000000 keys tested (404.56 k/s)
Time left: 2 days, 20 hours, 39 minutes, 25 seconds      0.01%
Current passphrase: 00006393

Master Key       : A4 F5 90 F5 BC 40 CA C5 4D 0A F6 BE 60 11 B8 44
                  C8 6F 12 EF FE 99 10 10 A3 3B 75 CE 53 78 D9 8B

Transient Key    : BF 57 FB FB AF A6 C9 5A 53 28 BB 08 24 1D 97 BE
                  B6 14 73 0B 07 AC 9B F1 30 85 E2 30 67 78 06 3B
                  A0 43 D3 75 18 ED BB CF 9B A8 DA C4 30 BF 72 16
                  04 04 AA 27 75 7E 6E 57 11 82 62 53 6A E3 23 49

EAPOL HMAC      : 69 1F 40 6E BC 3D 97 5B 95 11 A1 B7 DB F9 79 42
```

Figure 7. Aircrack-ng password-candidate testing against captured authentication data.

10. Wireless Information Observed

| Item                     | Observed Value    |
|--------------------------|-------------------|
| SSID                     | TP-Link_6922      |
| BSSID / Access Point MAC | B8:FB:B3:1C:69:22 |
| Channel                  | 2                 |
| Security                 | WPA2              |
| Cipher                   | CCMP              |
| Authentication           | PSK               |
| Wireless Interface       | wlan0             |

These details were recorded from the supplied project evidence.

11. Results and Findings

| Test                 | Observation                                                   | Status    |
|----------------------|---------------------------------------------------------------|-----------|
| Adapter detection    | Wi-Fi adapter and wlan0 were detected                         | Completed |
| Interface check      | wlan0 was visible                                             | Completed |
| Adapter information  | TP-Link adapter and driver were identified                    | Completed |
| Monitor mode         | Monitor mode was enabled                                      | Completed |
| Wireless scan        | Test network was visible                                      | Completed |
| Targeted capture     | Access point and client were visible                          | Completed |
| Wireless details     | SSID, BSSID, channel and security details were recorded       | Completed |
| Handshake screenshot | WPA handshake indicator clearly visible for B8:FB:B3:1C:69:22 | Completed |
| Password test        | Aircrack-ng testing and key-found evidence supplied           | Completed |

## Key Findings

### Finding 1 – Wi-Fi Adapter Worked

The USB Wi-Fi adapter was detected successfully and could be placed into monitor mode.

### Finding 2 – Wireless Information Was Visible

The test access point exposed basic information such as its SSID, BSSID, channel, and security type.

### Finding 3 – Client Activity Was Visible

The controlled client was visible during the targeted capture.

### Finding 4 – Password Strength Is Important

Password testing depends on the password candidates available in the wordlist. Captured authentication data does not automatically reveal the password.

### Finding 5 – Complete Evidence Is Important

The project now has evidence for scanning, monitoring, targeted capture, WPA/WPA2 handshake capture, and password testing. A separate wordlist screenshot may still be added if the evaluator requires it.

## 12. Password-Strength Test

Aircrack-ng was used to test password candidates against the authorized captured authentication data.

The supplied evidence shows Aircrack-ng running and testing password candidates. Another screenshot shows a KEY FOUND result.

The actual password is intentionally not included in this report.

This test shows why it is important to use a strong and unique Wi-Fi password. If a password is included in a candidate list, it may be identified through an offline dictionary-based test when the required authentication data is available.

## 13. Security Recommendations / Best Practices

1. Use **WPA3** if your router supports it.
2. Use a **long and strong Wi-Fi password**.
3. Do not use your name, phone number, birthday, or common words as the password.
4. Keep the **router software updated**.
5. Change the router's **default admin password**.
6. Turn off **WPS** if you do not need it.
7. Use a **separate guest network** for visitors and suitable smart devices.
8. Check the list of **connected devices** regularly.
9. Do Wi-Fi security testing only with **proper permission**.
10. Keep captured data safe and **do not share passwords or other private information**.

## 14. Troubleshooting

| Problem                     | Simple Solution                                                                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------|
| wlan0 is missing            | Reconnect the USB adapter and check the wireless interface.                                                         |
| Monitor mode does not start | Check the adapter and its driver.                                                                                   |
| Interface name changes      | Use the interface name currently shown by Kali Linux.                                                               |
| No networks appear          | Check monitor mode, adapter connection, driver, and distance from the test access point.                            |
| Client is not visible       | Make sure the authorized test client is connected.                                                                  |
| Handshake is not shown      | Keep the capture running while the controlled client reconnects.                                                    |
| Password is not found       | Check whether the test password is included in the authorized laboratory wordlist and whether the capture is valid. |

The commands in this report are documented for the authorized laboratory environment described in this project. The password-testing command uses placeholders for the approved wordlist and capture file.

## 15. Legal and Ethical Considerations

Wi-Fi security testing must always be performed with proper permission.

Before starting the test, the tester should receive permission from the network owner. Only the agreed network, devices, and testing period should be included in the testing.

Captured wireless information should be handled carefully and protected from unauthorized access.

Nearby networks belonging to other people should not be tested.

This project was performed as a controlled cybersecurity learning exercise using an authorized test environment.

## 16. Limitations

- The results depend on the Wi-Fi adapter and its driver.
- Different Wi-Fi adapters support different monitor-mode features.
- A dictionary-based test can only check passwords included in the wordlist.
- Capturing authentication data does not guarantee that the password will be recovered.
- Wireless signal conditions can affect packet capture.
- The project mainly covers basic WPA/WPA2 laboratory testing and does not cover every type of wireless attack.

## 17. Future Improvements

This project can be improved by learning more about different areas of Wi-Fi security. In the future, I would like to study **WPA3 security** to understand how newer Wi-Fi networks protect user data and authentication.

I would also like to learn about **wireless intrusion detection**, which helps identify suspicious activity on a Wi-Fi network. Another area is **rogue access point detection**, which can help find unauthorized Wi-Fi devices connected to or operating near a network.

The project can also be extended to study **enterprise Wi-Fi authentication**, which is commonly used in organizations. **Automated wireless security-audit reports** could make it easier to collect test results and prepare reports. Finally, learning about **network segmentation** and **guest-network security** would help improve how different devices and users are separated and protected on a Wi-Fi network.

Overall, these improvements would help me understand Wi-Fi security in more detail and build better skills for future cybersecurity projects.

## 18. Conclusion

This project helped me understand the basic process of performing a Wi-Fi security audit using Kali Linux and Aircrack-ng.

During the project, I learned how to check a USB Wi-Fi adapter, identify the wireless interface, enable monitor mode, scan wireless networks, record wireless information, and perform a targeted capture.

I also learned the basic purpose of WPA/WPA2 authentication data and how it can be used in a controlled password-strength test. The supplied evidence shows the Aircrack-ng password test and a key-found result.

The project also helped me understand why strong Wi-Fi passwords, updated router software, and proper network configuration are important.

The main lesson from this project is that wireless security testing must always be performed with permission and only within the approved testing scope.

## 19. References

The report uses the following references:

- **Kali Linux documentation.**  
<https://www.kali.org/docs/>
- **Aircrack-ng documentation.**  
<https://www.aircrack-ng.org/doku.php?id=airodump-ng>
- **IEEE 802.11 wireless LAN standards.**  
<https://standards.ieee.org/ieee/802.11/10548/>
- **Wi-Fi Alliance security guidance.**  
<https://www.wi-fi.org/security>

## Practical Steps

```
iwconfig
```

**Check wireless interface**

```
ip link
```

**Check network interfaces**

```
sudo airmon-ng
```

### Check Aircrack-ng adapter information

```
sudo airmon-ng start wlan0
```

### Enable monitor mode

```
sudo airodump-ng wlan0
```

### Scan wireless networks

```
sudo airodump-ng --bssid B8:FB:B3:1C:69:22 --channel 2 wlan0
```

### Target the authorized access point

```
sudo aircrack-ng -w <AUTHORIZED_WORDLIST> <CAPTURE_FILE>.cap
```

### Authorized password-strength test

## Appendix A – Assignment Requirement Check

The project requirements were compared with the completed report.

| Assignment Requirement        | Status   |
|-------------------------------|----------|
| Scanning and monitoring setup | Covered  |
| Handshake capture             | Covered  |
| Password-cracking attempt     | Covered  |
| Findings documentation        | Covered  |
| Screenshots                   | Included |
| Best-practices guide          | Covered  |

The supplied evidence supports adapter detection, monitor mode, wireless scanning, targeted capture, and Aircrack-ng password testing.

#### Proof 1 – Wi-Fi Adapter Detection

Shows that the wlan0 wireless interface and TP-Link adapter were detected.

#### Proof 2 – Interface Check

Shows that the wlan0 wireless interface was available.

#### Proof 3 – Adapter Information

Shows the TP-Link adapter, driver, and chipset information.

#### Proof 4 – Monitor Mode

Shows that monitor mode was enabled.

#### Proof 5 – Wireless Scan and Client Observation

Shows the test access point, wireless security information, and visible wireless stations.

#### Proof 6 – Targeted Packet Capture

Shows the selected test access point and client stations during targeted capture.

#### Proof 7 – Aircrack-ng Password Testing

Shows Aircrack-ng testing password candidates.

### **Proof 8 – WPA Handshake Capture**

Shows the WPA handshake indicator for the authorized test network TP-Link\_6922, with BSSID B8:FB:B3:1C:69:22 on channel 2.

### **Proof 9 – Key-Found Result**

Shows the KEY FOUND result from the supplied Aircrack-ng evidence. The actual password is not reproduced in the report.

## **Ethical and Legal Practice**

Wi-Fi security testing should only be performed after getting permission from the network owner. Before starting the testing, the tester should clearly decide which network, devices, testing time, and activities are allowed.

The testing should be limited to the approved scope. Other nearby Wi-Fi networks should not be scanned or tested without permission. Any captured wireless information should also be stored safely and should not be shared with unauthorized people.

In this project, the testing was performed in a controlled environment on an authorized test network. The main purpose was to learn Wi-Fi security testing while following responsible and ethical practices.